



COURSE PLAN

Academic Session	: 2025-2026(Even Semester)	Semester	: 02-2025 Sem 2
Course	: BTRCS	Subject Code	: 4012UCS
Branch	: 0101BTR15	Name of the Subject	: Computer Organization and Arch
Groups	: DS4A	Name of Faculty Member	: Abhishek .
Contact	: 0	Location	:

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Unit No. & Description	Topics	Text Book No.	Chapter & Page No.	No. of Proposed Lectures	Teaching/Learning Methodology
Unit-01 CIRCUIT DESIGN	Arithmetic algorithms (addition, subtraction, Boot	[T1]	3(74-80)	1	Lecture/Discussion
	Bus and Memory transfers,	[T1]	4(97-99)	1	Lecture/Discussion
	IEEE754 standard for Floating point numbers	[T1]	3(83)	1	Lecture/Discussion
	Register transfer language,	[T1]	4(93-96)	1	Lecture/Discussion
	Tristate and multiplexed bus architecture	[T1]	4(102-110)	1	Lecture/Discussion
	arithmetic, logic & shift micro Operations	[T1]	6(179-181)	1	Lecture/Discussion
	assembly & machine language	[T1]	6(179-181)	1	Lecture/Discussion
Unit-02 CONTROL DESIGN	Execution of branch instructions, Execution of	[T1]	5(147-150)	1	Lecture/Discussion
	Fetching a word from memory, storing a word in mem	[T1]	5(147-150)	1	Lecture/Discussion
	Hardwired: Fundamental Concepts: Register	[T1]	8(242-243)	1	Lecture/Discussion
	Micro Programmed: Microinstruction, Micro program	[T1]	7(216-220)	1	Lecture/Discussion
	Microinstruction with Next-address field,	[T1]	2(222-225)	1	Lecture/Discussion
	Performing of arithmetic or logical operations,	[T1]	6(192-196)	1	Lecture/Discussion
	Pipelining.	[T1]	9(302)	1	Lecture/Discussion
	Wide-Branch addressing, Pre fetching Microinstruct	[T1]	7(216)	1	Lecture/Discussion
Unit-03 PROCESSOR DESIGN	Addressing mode	[T1]	8(260-266)	1	Lecture/Discussion
	Data transfer & manipulations,	[T1]	8(266-267)	1	Lecture/Discussion
	Instruction format (zero address, 1 address, 2 add	[T1]	8(255-259)	1	Lecture/Discussion
	Micro instructions,	[T1]	7(222-224)	1	Lecture/Discussion
	Processor Organization: General register organizat	[T1]	8(242-244)	1	Lecture/Discussion

	Program Control	[T1]	8(273-275)	1	Lecture/Discussion
	Reduced Instruction Set Computer.	[T1]	8(282-285)	1	Lecture/Discussion
Unit-04 INPUT-OUTPUT ORGANIZATION	Asynchronous Parallel Communication (Strobe Contro	[T1]	11(391-395)	1	Lecture/Discussion
	Asynchronous Serial Communication	[T1]	11(396-400)	1	Lecture/Discussion
	Direct Memory Access,	[T1]	11(415-418)	1	Lecture/Discussion
	I/O devices and Interface	[T1]	11(415-418)	1	Lecture/Discussion
	Input-Output Processor,	[T1]	11(385-386)	1	Lecture/Discussion
	Interrupts, their types & Interrupt handling,	[T1]	11(407-411)	1	Lecture/Discussion
	Stack organization,	[T1]	8(247-249)	1	Lecture/Discussion
Unit-05 MEMORY ORGANISATION	Cache Memories, Performance Considerations,	[T1]	12(462-468)	1	Lecture/Discussion
	Cache memory mapping, Auxiliary memory,	[T1]	12(452-455)	1	Lecture/Discussion
	Main Memory (RAM/ROM),	[T1]	12(452-455)	1	Lecture/Discussion
	Memory Hierarchy, Locality, inclusion & coherence	[T1]	12(448-449)	1	Lecture/Discussion
	Memory management, Secondary Storage.	[T1]	12(448-449)	1	Lecture/Discussion
	Modes of transfer,	[T1]	12(450)	1	Lecture/Discussion
	Speed Size and Cost	[T1]	12(465)	1	Lecture/Discussion
	Synchronous Serial Communication,	[T1]	12(480)	1	Lecture/Discussion
	Virtual Memory (paging and segmentation), Page fau	[T1]	11(396)	1	Lecture/Discussion
	ALU design	[T1]	11(396)	1	Lecture/Discussion

Note: Sequence of teaching unit- 01 02 03 04 05

Teacher's Assessment:

Name of the Activity

Marks Allotted to Each Activity

Date of Completion

No. of Proposed Lectures

40

Activities:

Session Allotted

Details of other Activities Planned:

References :

Text Books :

T1

Computer System Architecture,M. Mano, , ,

Reference Books :

Other References :

Abbreviations:

L : Lecture

C : Credit

(Signature of Faculty Member with date)

(Signature of Dean with date)